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The Feynman Rules

In previous chapters the use of covariant free fields in the construction of the Hamiltonian density has been motivated by the requirement that the S -matrix satisfy Lorentz invariance and cluster decomposition conditions. With the Hamiltonian density constructed in this way, it makes no difference which form of perturbation theory we use to calculate the S -matrix; the results will automatically satisfy these invariance and clustering conditions in each order in the interaction density. Nevertheless, there are obvious practical advantages in using a version of perturbation theory in which the Lorentz invariance and cluster decomposition properties of the S -matrix are kept manifest at every stage in the calculation. This was not true for the perturbation theory used in the 1930s, now known as ‘old-fashioned perturbation theory’, described at the beginning of Section 3.5. The great achievement of Feynman, Schwinger, and Tomonaga in the late 1940s was to develop perturbative techniques for calculating the S -matrix, in which Lorentz invariance and cluster decomposition properties are transparent throughout. This chapter will outline the diagrammatic calculational technique first described by Feynman at the Poconos Conference in 1948. Feynman was led to these diagrammatic rules in part through his development of a path-integral approach, which will be the subject of Chapter 9. In this chapter, we shall use the approach described by Dyson¹ in 1949, which until the 1970s was the basis of almost all analyses of perturbation theory in quantum field theory, and still provides a particularly transparent introduction to the Feynman rules.

6.1 Derivation of the Rules

Our starting point is a formula for the S -matrix, obtained by putting together the Dyson series (3.5.10) with expression (4.2.2) for the free-particle states: